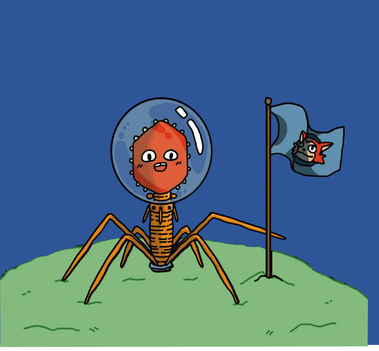

SpacePHARER: Sensitive identification of phages from the CRISPR spacers in their prokaryotic hosts

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Abstract

Phages (bacteria- and archaea-infecting viruses) are key components in most ecosystems, and crucial for understanding microbial dynamics. Despite the increasing pace of discovering new phages, enabled by viral metagenomics, identifying their host remains a challenge. Several computational approaches to predict phage-host relationships utilize CRISPR spacers [1,2]. These are short sequences (28-42nt) acquired by prokaryotes during phage infection as a defense mechanism. The traditional method taking this approach searches with BLASTN for matches to spacers in phage genomes, which is limited by its low sensitivity and low speed. **SpacePHARER** (CRISPR **Spacer** **P**hage-**H**ost **P**air Find**er**) [3] is a sensitive and fast tool for *de novo* prediction of phage-host relationships via identifying phage genomes that match CRISPR spacers in genomic or metagenomic data.

Why is SpacePHARER sensitive?

Six-frame translated search: phage genomes are mostly coding, and mutations should cause minimal changes in the AA sequence

Short sequence search: optimized search parameters for matching very short sequences

Combine numerous (weak) signals: from multiple spacers matching to the same phage

Availability

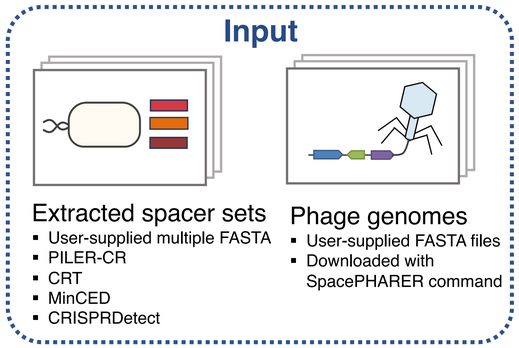
SpacePHARER is free open-source software (GPLv3)

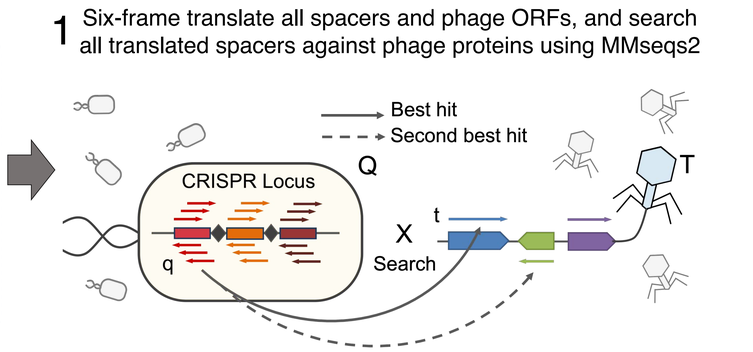
spacepharer.soedinglab.org

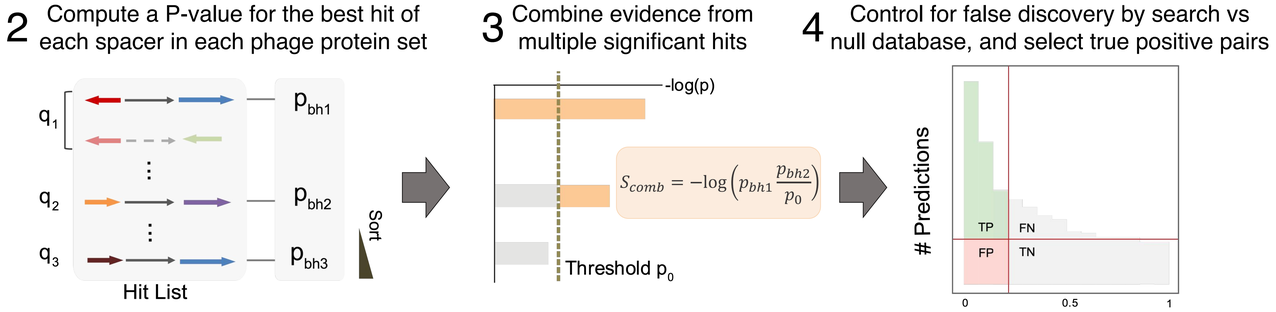
Install from Bioconda:

conda install -c conda-forge -c bioconda spacepharer

Method



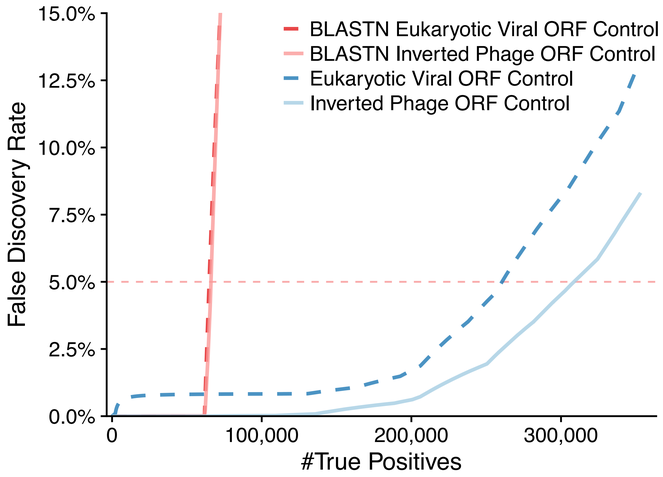




Performance

We evaluated the performance by searching all unique spacers from 24,322 prokaryotic genomes [2] against 7,824 phage genomes downloaded from NCBI. We used inverted phage sequences (solid lines) or eukaryotic viral ORFs (dashed lines) as null database.
At a false discovery rate (FDR) of 0.05, SpacePHARER predicted 319,029 (inverted phage sequences as null) and 253,419 (eukaryotic viral ORFs as null) prokaryote-phage matches, 4x more sensitive than BLASTN in detecting phage-host pairs.

Runtime: 12 mins (2×6-core 2.40 GHz) (speed up of >13x over BLASTN)



Conclusion

- SpacePHARER *de novo* identifies host-phage pairs by matching CRISPR-Cas spacer sequences to the phage genomes
- SpacePHARER improves the sensitivity of protospacer search on multiple levels and identifies 4x host-phage pairs than BLASTN
- SpacePHARER is fast and efficient enough to analyze large-scale genomic and metagenomic datasets, as the spacer and phage databases are both growing rapidly

References

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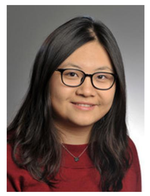
Contact



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